Week 6 - Exercises

For this week's tutorial as for the previous ones, we will use data from Bruce Hansen's econometric textbook which can be downloaded here.

Exercise 1

Consider the dataset `AJR2001` used in exercise 3 of week5. Remember that the authors wanted to estimate the causal effect of good institutions (measured by the variable *risk*) on economic development (measured by log (GDP per capita$_i$)). For that they instrument, the variable risk by the log of the mortality rate of the settlers.

1. Re-estimate the model by 2SLS making sure that the standard errors you get are robust to heteroskedasticity (make sure to use robust standard errors in the rest of the exercise).

Solution:

```
    . ivregress 2sls loggdp (risk=logmort0 ), robust

Instrumental variables 2SLS regression            Number of obs   =         64
                                                  Wald chi2(1)    =      29.86
                                                  Prob > chi2     =     0.0000
                                                  R-squared       =     0.1880
                                                  Root MSE        =     .93672

------------------------------------------------------------------------------
             |               Robust
      loggdp | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        risk |   .9294897   .1700872     5.46   0.000     .5961249    1.262855
       _cons |   1.994296   1.134985     1.76   0.079    -.2302345    4.218826
------------------------------------------------------------------------------
Instrumented: risk
Instruments: logmort0
```

The robust command reports heteroskedasticity-robust standard errors.

2. Using two different Stata commands to show that the relevance assumption of the instrument is satisfied.

Solution:

The first way is to perform the F-test on the instrument yourself to verify that the instrument is sufficiently correlated with the regressor. To do that you first need to regress the endogenous variable on the instrument and then perform an F-test on the coefficient of the instrument (as we have just one parameter, testing the linear restriction here is similar to testing for significance except that we do not use the t-stat for this here but the F-stat). We have mentionned that as a rule-of-thumb the F-statistics from this test should be 10 or more.

Below you can see that the F-stat is 16.33 which is larger than 10, so our instrument seems to be relevant (make sure to know the difference between relevance and validity). You can also see that the p-value is 0 meaning that it is very unlikely to obtain such a value if this F-stat was following the distribution F(1,62) (which it should follow under the null that the coefficient on `logmort0` *is zero).*

```
reg risk logmort0, robust

Linear regression                               Number of obs     =         64
                                                F(1, 62)          =      16.33
                                                Prob > F          =     0.0001
                                                R-squared         =     0.2735
                                                Root MSE          =      1.262

------------------------------------------------------------------------------
             |               Robust
        risk | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    logmort0 |  -.6132892   .1517849    -4.04   0.000     -.916703   -.3098755
       _cons |   9.365895   .7084137    13.22   0.000     7.949796    10.78199
------------------------------------------------------------------------------

. test logmort0

 ( 1)  logmort0 = 0

       F(  1,    62) =   16.33
            Prob > F =    0.0001
```

The second way is to use the Stata command firststage after ivregress. You can see that the reported F-stat is the same as above.

```
    . ivregress 2sls loggdp (risk=logmort0 ), robust

Instrumental variables 2SLS regression            Number of obs   =         64
                                                  Wald chi2(1)    =      29.86
                                                  Prob > chi2     =     0.0000
                                                  R-squared       =     0.1880
                                                  Root MSE        =     .93672

------------------------------------------------------------------------------
             |               Robust
      loggdp | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        risk |   .9294897   .1700872     5.46   0.000     .5961249    1.262855
       _cons |   1.994296   1.134985     1.76   0.079    -.2302345    4.218826
------------------------------------------------------------------------------
Instrumented: risk
 Instruments: logmort0

. estat firststage

  First-stage regression summary statistics
  -----------------------------------------------------------------------------
               |            Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.       F(1,62)   Prob > F
  -------------+---------------------------------------------------------------
          risk |  0.2735      0.2618       0.2735       16.3258    0.0001
  -----------------------------------------------------------------------------
```

3. Consider now using as an instrument not the log of the mortality rate but its level. Would you recommend using this variable as an instrument here?

Solution:

If log mortality is a valid instrument, it makes sense that mortality should be a valid instrument as well (i.e. uncorrelated with the potential omitted variables creating endogeneity issues and only affecting log gdp per capita through its effect on the variable risk.)

But we need to also check whether mortality is relevant. Doing the same test as in the previous question, we see that mortality is a worse predictor of risk than log mortality. In particular the F-stat for mortality is 7.02 which is lower than 10.

Log mortality seems therefore more recommended as an instrument than mortality.

```
    . gen mortality = exp(logmort0)

. reg risk mortality, robust

Linear regression                               Number of obs     =         64
                                                F(1, 62)          =       7.02
                                                Prob > F          =     0.0102
                                                R-squared         =     0.0640
                                                Root MSE          =     1.4325

------------------------------------------------------------------------------
             |               Robust
        risk | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   mortality |  -.0007862   .0002967    -2.65   0.010    -.0013792   -.0001931
       _cons |   6.709419   .2010414    33.37   0.000     6.307543    7.111295
------------------------------------------------------------------------------

. test mortality

 ( 1)  mortality = 0

       F(  1,    62) =    7.02
            Prob > F =    0.0102

.
. ivregress 2sls loggdp (risk=mortality), robust

Instrumental variables 2SLS regression            Number of obs   =         64
                                                  Wald chi2(1)    =       5.56
                                                  Prob > chi2     =     0.0183
                                                  R-squared       =          .
                                                  Root MSE        =     1.1817

------------------------------------------------------------------------------
```

```
             |               Robust
      loggdp | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        risk |   1.160549   .4920554     2.36   0.018     .1961384     2.12496
       _cons |   .4886904   3.230538     0.15   0.880    -5.843049    6.820429
------------------------------------------------------------------------------
Instrumented: risk
 Instruments: mortality

. estat firststage

  First-stage regression summary statistics
  --------------------------------------------------------------------------
               |            Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.       F(1,62)   Prob > F
  -------------+------------------------------------------------------------
         risk |  0.0640      0.0489       0.0640       7.02272    0.0102
  --------------------------------------------------------------------------
```

4. Using two different ways to check whether the variable risk is endogenous.

Solution:

A first way is to perform the Hausman test. The procedure here is as follow:

1. *regress endogenous regressor on instrument (i.e. 1st stage of 2SLS)*
2. *compute the residual from this regression*
3. *regress dependent variable on endogenous regressor and the previously computed residual (i.e. one of the 2 ways to perform the 2nd stage of 2SLS - see previous tutorial)*
4. *do an F-test on the coefficient of the residual*

The last step checks whether the coefficient on the residual is significant. If it is the case, then it indicates that 2SLS and OLS will deliver statistically significantly different results.

If our instrument is valid, then the OLS estimator should be consistent only if exogeneity is satisfied, while the 2SLS estimator should be consistent both under exogeneity and under endogeneity. Therefore, under exogeneity

both OLS and 2SLS should converge in probability to the same value, while under endogeneity they should converge in probability to different values (with the 2SLS being the estimator which is consistent if the instrument is valid).

Here, we see that the p-value of our F-test is 0 so we can safely reject the null that the residual is insignificant in the second-stage regression. Therefore, we can safely reject exogeneity. So the 2SLS estimator should be consistent, while OLS will not be.

```
    . reg risk logmort0, robust

Linear regression                                      Number of obs     =         64
                                                       F(1, 62)          =      16.33
                                                       Prob > F          =     0.0001
                                                       R-squared         =     0.2735
                                                       Root MSE          =      1.262

------------------------------------------------------------------------------
             |               Robust
        risk | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    logmort0 |  -.6132892   .1517849    -4.04   0.000     -.916703   -.3098755
       _cons |   9.365895   .7084137    13.22   0.000     7.949796    10.78199
------------------------------------------------------------------------------

. predict risk_pred
(option xb assumed; fitted values)

. predict risk_err, residual

. reg loggdp risk risk_err, robust

Linear regression                                      Number of obs     =         64
                                                       F(2, 61)          =     114.48
                                                       Prob > F          =     0.0000
                                                       R-squared         =     0.6501
                                                       Root MSE          =     .62982

------------------------------------------------------------------------------
             |               Robust
      loggdp | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        risk |   .9294897    .088528    10.50   0.000      .752467    1.106512
```

```
    risk_err |     -.5689   .1234492    -4.61   0.000    -.8157518   -.3220481
       _cons |   1.994296   .5620951     3.55   0.001      .870317    3.118274
------------------------------------------------------------------------------

. test risk_err

 ( 1)  risk_err = 0

       F(  1,    61) =   21.24
            Prob > F =    0.0000

.
end of do-file
```

An alternative way is to use the command estat endogenous after ivregress which reports the same F-stat. The first way hopefully helps you better understand what we are testing and how.

```
    . ivregress 2sls loggdp (risk=logmort0 ), robust

    Instrumental variables 2SLS regression            Number of obs   =         64
                                                      Wald chi2(1)    =      29.86
                                                      Prob > chi2     =     0.0000
                                                      R-squared       =     0.1880
                                                      Root MSE        =     .93672

    ------------------------------------------------------------------------------
                 |               Robust
          loggdp | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
    -------------+----------------------------------------------------------------
            risk |   .9294897   .1700872     5.46   0.000     .5961249    1.262855
           _cons |   1.994296   1.134985     1.76   0.079    -.2302345    4.218826
    ------------------------------------------------------------------------------
    Instrumented: risk
     Instruments: logmort0

    . estat endogenous

      Tests of endogeneity
      H0: Variables are exogenous
```

```
Robust score chi2(1)            =  9.08197  (p = 0.0026)
Robust regression F(1,61)       =  21.2371  (p = 0.0000)
```

Exercise 2

This exercises uses the dataset `Card1995`. The aim of Card was to estimate the return to schooling by estimating linear model of the form:

$$\log wage_i = \beta_0 + \beta_1 education_i + \beta_2 exper_i + \beta_3 exper_i^2 + \mathbf{x}_i'\gamma + \varepsilon_i$$

where $education_i$ is the number of years of education and $exper_i$ is the number of years of experience approximated by $age_i - education_i - 6$. $\mathbf{x}_i$ is a vector of controls. For this exercise use only observations for which the $\log wage_i$ is observable.

1. Why estimating such equation by OLS would be problematic for inferring the returns to schooling? Show nonetheless the results of such regression (without controls). How would you interpret the coefficients?

Solution: *It is problematic because other factors, such as ability may affect both wage and education. Therefore, the correlation between wage and education that OLS will capture is likely not to be (completely) causal. In other words, such omitted variable creates the possibility of endogeneity and in such case OLS will be inconsistent for the causal effect we are after.*

Importantly, when interpreting the results from such an OLS regression we should careful and talk about statistical association but not about causality. In particular, the regression below shows that one more year of education is associated with a wage which is 9.3% higher (after controlling for experience). However, it does NOT show (if we think the endogeneity issue to be a serious concern) that pursuing one more year of education increases wage by 9.3%. Such statement assumes causality when there is a serious concern about omitted variable bias. Be careful in how you interpret this kind of results. Finally, for experience we see that the quadradic relation we assumed seem justified from the results we get.

```
    . drop if missing(lwage76)
(603 observations deleted)


.
. * 1* regress OLS
. gen exper  = age76 - ed76 - 6

. gen exper2 = exper^2 / 100
```

```
. rename age76 age

. gen age2 = age^2

.
. reg lwage76 ed76 exper exper2, robust

Linear regression                               Number of obs     =      3,010
                                                F(3, 3006)        =     231.75
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1958
                                                Root MSE          =     .39818

------------------------------------------------------------------------------
             |               Robust
     lwage76 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ed76 |   .0931707   .0036791    25.32   0.000     .0859568    .1003846
       exper |   .0897828   .0070827    12.68   0.000     .0758953    .1036703
      exper2 |  -.2485938   .0340813    -7.29   0.000    -.3154188   -.1817687
       _cons |    4.46854   .0702761    63.59   0.000     4.330746    4.606334
------------------------------------------------------------------------------

.
end of do-file

.
```

2. Card argues that, if a potential student lives close to a college, it lowers the cost of going to college. At the same time, he argues that living close to a college has plausibly no direct impact on wage. He therefore consider as an instrument the variable `nearc4` which indicates whether person i grew up close to a 4-year college. What other condition should you have for college proximity to be a valid instrument?

Solution:

We need in addition to have the instrument, here `nearc4`*, uncorrelated with the omitted variables which may generate the omitted variable bias we are trying to solve with instrumental variable methods. For instance, we*

would like `nearc4` *to be uncorrelated with variables such as ability when we also control for the other regressors in our regression.*

3. Assume for now that the latter condition is satisfied. Use `nearc4` as instrument to estimate the equation of interest (again without controls) and check that the relevance assumption of the instrument is satisfied.

Solution:

Here, I want to highlight how the test for relevance "differs" from the one of the previous exercise. Here, unlike in the previous exercise, we have control variables. So if you test for relevance you need to test for relevance while controlling for the exogenous regressors. This is what is done below when I show how to do the test without using the firststage command in stata. Our F-stat is way larger than 10 here. So our instrument seems relevant (although remember nothing proves it is valid).

One thing to notice here is that the coefficient on education is significantly higher in the 2SLS regression than in the OLS case with point estimates of .25 and .09 respectively, and 95% confidence intervals that are far away from one another (this is from that that we can infer that the difference is significant from a statistical viewpoint).

```
    . ivregress 2sls lwage76  exper exper2 (ed76  = nearc4), robust

Instrumental variables 2SLS regression            Number of obs   =      3,010
                                                  Wald chi2(3)    =      89.98
                                                  Prob > chi2     =     0.0000
                                                  R-squared       =          .
                                                  Root MSE        =     .52053

------------------------------------------------------------------------------
             |               Robust
     lwage76 | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ed76 |   .2587156   .0337394     7.67   0.000     .1925875    .3248436
       exper |   .1596791   .0169521     9.42   0.000     .1264537    .1929045
      exper2 |  -.2487532   .0473575    -5.25   0.000    -.3415722   -.1559341
       _cons |   1.653984   .5742697     2.88   0.004     .5284365    2.779532
------------------------------------------------------------------------------
Instrumented: ed76
 Instruments: exper exper2 nearc4

. estat firststage
```

```
  First-stage regression summary statistics
  --------------------------------------------------------------------------
               |            Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.     F(1,3006)   Prob > F
  -------------+------------------------------------------------------------
          ed76 |  0.4372      0.4367       0.0189       58.9916    0.0000
  --------------------------------------------------------------------------



.
. reg ed76 nearc4 exper exper2  , robust

Linear regression                                      Number of obs     =      3,010
                                                       F(3, 3006)        =    1072.66
                                                       Prob > F          =     0.0000
                                                       R-squared         =     0.4372
                                                       Root MSE          =     2.0092

------------------------------------------------------------------------------
             |               Robust
        ed76 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      nearc4 |   .6002325   .0781492     7.68   0.000     .4470012    .7534638
       exper |  -.4225143   .0325957   -12.96   0.000    -.4864264   -.3586021
      exper2 |   .0235024   .1735007     0.14   0.892    -.3166896    .3636944
       _cons |   16.57345   .1407528   117.75   0.000     16.29746    16.84943
------------------------------------------------------------------------------

. test nearc4

 ( 1)  nearc4 = 0

       F(  1,  3006) =   58.99
            Prob > F =    0.0000


.
end of do-file


.
```

4. For a subset of the men in the sample, an IQ measure is available. Regress IQ on `nearc4`. Why should the results from this regression concern you regarding the results you obtained in 3?

Solution:

We see that iq is possitively related to college proximity. This casts doubt on the validity of our instrument. Indeed, to the extent that IQ captures (at least partly) ability, we want our instrument to be uncorrelated with it. This is clearly not the case here. So our instrumental variable estimation may reproduce the same kind of endogeneity issue that we were trying to solve.

```
. reg iq nearc4, robust

Linear regression                               Number of obs     =      2,061
                                                F(1, 2059)        =      12.00
                                                Prob > F          =     0.0005
                                                R-squared         =     0.0059
                                                Root MSE          =     15.382

------------------------------------------------------------------------------
             |               Robust
          iq | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      nearc4 |     2.5962   .7494531     3.46   0.001     1.126435    4.065965
       _cons |   100.6106   .6331052   158.92   0.000     99.36906    101.8522
------------------------------------------------------------------------------
```

5. Now regress IQ on `nearc4`, the regional dummies `reg661` to `reg669` and the south indicator `smsa66`. What do you see? What would you then recommend as an adaptation to the estimation you did in 3 if you want to use the full sample at your disposition (and not just those observations for which IQ is observable)? Implement it.

Solution:

We see that IQ is no longer correlated with college proximity once we control for those additional factors. This suggest that our instrument may actually be valid but ONLY if we control for such geographical factors in our regression. This is important to keep in mind. An instrument may be valid if you control for enough relevant factors, but not valid if you do not use sufficient controls. When controlling for the factors here we are in fact assuming that college proximity is unrelated to ability (and other omitted factors of concerns) WITHIN a region. This is weaker

than assuming that assuming that college proximity is unrelated to ability at the level of the U.S. (as some regions may have more colleges and on average people with higher IQs).

We implement this below. We can remark two things. First, our instrument is still relevant but our F-stat is much lower than before (at about 12 now). Second, our point estimate on education is lower (in line with the fact that our previously invalid 2SLS was partly capturing differences in IQ/ability) and importantly our 2SLS and OLS confidence intervals now overlap. This is because of two things: 1) our point estimate is lower and 2) the 2SLS estimator is less precise (i.e. the confidence interval widens) probably due to the fact that our instrument explains less of the variations of our endogenous regressor.

```
    . reg iq nearc4 reg66* smsa66, robust
note: reg661 omitted because of collinearity.

Linear regression                                      Number of obs     =      2,061
                                                       F(10, 2050)       =      12.63
                                                       Prob > F          =     0.0000
                                                       R-squared         =     0.0626
                                                       Root MSE          =     14.969

------------------------------------------------------------------------------
             |               Robust
          iq | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      nearc4 |   .3478974    .807889     0.43   0.667    -1.236471    1.932266
      reg661 |          0  (omitted)
      reg662 |   1.099282   1.551963     0.71   0.479    -1.944306    4.142871
      reg663 |  -1.559295   1.509444    -1.03   0.302    -4.519499    1.400908
      reg664 |  -.5425011   1.723682    -0.31   0.753    -3.922852     2.83785
      reg665 |   -8.47546   1.619462    -5.23   0.000    -11.65142   -5.299497
      reg666 |  -7.421172    1.93497    -3.84   0.000    -11.21588    -3.62646
      reg667 |   -8.39441   1.795532    -4.68   0.000    -11.91567   -4.873152
      reg668 |  -2.924975   2.338128    -1.25   0.211    -7.510329    1.660379
      reg669 |  -2.891917   1.722389    -1.68   0.093    -6.269732    .4858982
     smsa66r |   1.089165   .7947269     1.37   0.171    -.4693908    2.647722
       _cons |   104.7735   1.505288    69.60   0.000     101.8214    107.7255
------------------------------------------------------------------------------


    . ivregress 2sls lwage76  exper exper2 reg66* smsa66  (ed76  = nearc4 ), robust
note: reg669 omitted because of collinearity.

Instrumental variables 2SLS regression            Number of obs   =      3,010
```

```
                                                  Wald chi2(12)   =     458.94
                                                  Prob > chi2     =     0.0000
                                                  R-squared       =     0.1866
                                                  Root MSE        =     .40018

------------------------------------------------------------------------------
             |               Robust
     lwage76 | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ed76 |   .1416411    .058318     2.43   0.015     .0273399    .2559424
       exper |   .1114847   .0258525     4.31   0.000     .0608147    .1621547
      exper2 |  -.2427267   .0359691    -6.75   0.000    -.3132249   -.1722285
      reg661 |  -.1095306   .0422102    -2.59   0.009    -.1922611   -.0268001
      reg662 |  -.0061608   .0345057    -0.18   0.858    -.0737907     .061469
      reg663 |   .0302294   .0347417     0.87   0.384     -.037863    .0983219
      reg664 |  -.0648004   .0397716    -1.63   0.103    -.1427514    .0131505
      reg665 |   -.132501   .0615952    -2.15   0.031    -.2532253   -.0117766
      reg666 |  -.1155272   .0666407    -1.73   0.083    -.2461405     .015086
      reg667 |  -.1409766   .0577312    -2.44   0.015    -.2541276   -.0278255
      reg668 |  -.1946134   .0525725    -3.70   0.000    -.2976536   -.0915731
      reg669 |          0  (omitted)
     smsa66r |   .0803322   .0281337     2.86   0.004     .0251911    .1354733
       _cons |   3.639692   1.001967     3.63   0.000     1.675874     5.60351
------------------------------------------------------------------------------
Instrumented: ed76
 Instruments: exper exper2 reg661 reg662 reg663 reg664 reg665 reg666 reg667
              reg668 smsa66r nearc4

. estat firststage

  First-stage regression summary statistics
  -----------------------------------------------------------------------
               |             Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.     F(1,2997)   Prob > F
  -------------+---------------------------------------------------------
          ed76 |  0.4578      0.4557       0.0039       12.3619    0.0004
  -----------------------------------------------------------------------



.
```

```
. reg ed76 nearc4 exper exper2  reg66* smsa66 , robust
note: reg668 omitted because of collinearity.

Linear regression                                      Number of obs     =      3,010
                                                       F(12, 2997)       =     285.64
                                                       Prob > F          =     0.0000
                                                       R-squared         =     0.4578
                                                       Root MSE          =      1.975

------------------------------------------------------------------------------
             |               Robust
        ed76 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      nearc4 |   .3032451   .0862486     3.52   0.000     .1341327    .4723575
       exper |  -.4257313   .0319351   -13.33   0.000    -.4883482   -.3631144
      exper2 |   .0885551   .1691419     0.52   0.601    -.2430909    .4202011
      reg661 |  -.5278016   .2551155    -2.07   0.039    -1.028021   -.0275824
      reg662 |  -.6211287   .2196758    -2.83   0.005    -1.051859    -.190398
      reg663 |  -.6251041   .2130414    -2.93   0.003    -1.042826   -.2073819
      reg664 |  -.4388472   .2381626    -1.84   0.065    -.9058259    .0281315
      reg665 |  -1.221944    .212881    -5.74   0.000    -1.639352   -.8045368
      reg666 |  -1.278713   .2270927    -5.63   0.000    -1.723986   -.8334395
      reg667 |  -1.102944   .2240033    -4.92   0.000     -1.54216   -.6637283
      reg668 |          0  (omitted)
      reg669 |  -.3348251   .2315281    -1.45   0.148    -.7887951    .1191448
     smsa66r |   .2735886   .0880339     3.11   0.002     .1009756    .4462016
       _cons |   17.36837   .2406549    72.17   0.000      16.8965    17.84023
------------------------------------------------------------------------------

. test nearc4

 ( 1)  nearc4 = 0

       F(  1,  2997) =   12.36
            Prob > F =    0.0004
```

6. Do an endogeneity test and comment on it.

Solution:

We can no longer reject exogeneity (once again see how you can do this test without using the estat endogenous command). This is in line with what we mentionned above about the fact that the OLS and 2SLS confidence intervals now overlap.

```
* after ivregress
    . estat endogenous

  Tests of endogeneity
  H0: Variables are exogenous

  Robust score chi2(1)            =  1.11924  (p = 0.2901)
  Robust regression F(1,2996)     =  1.11339  (p = 0.2914)


  . reg ed76 nearc4 exper exper2  reg66* smsa66 , robust
note: reg668 omitted because of collinearity.

Linear regression                               Number of obs     =      3,010
                                                F(12, 2997)       =     285.64
                                                Prob > F          =     0.0000
                                                R-squared         =     0.4578
                                                Root MSE          =      1.975

------------------------------------------------------------------------------
             |               Robust
        ed76 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      nearc4 |   .3032451   .0862486     3.52   0.000     .1341327    .4723575
       exper |  -.4257313   .0319351   -13.33   0.000    -.4883482   -.3631144
      exper2 |   .0885551   .1691419     0.52   0.601    -.2430909    .4202011
      reg661 |  -.5278016   .2551155    -2.07   0.039    -1.028021   -.0275824
      reg662 |  -.6211287   .2196758    -2.83   0.005    -1.051859    -.190398
      reg663 |  -.6251041   .2130414    -2.93   0.003    -1.042826   -.2073819
      reg664 |  -.4388472   .2381626    -1.84   0.065    -.9058259    .0281315
      reg665 |  -1.221944    .212881    -5.74   0.000    -1.639352   -.8045368
      reg666 |  -1.278713   .2270927    -5.63   0.000    -1.723986   -.8334395
      reg667 |  -1.102944   .2240033    -4.92   0.000     -1.54216   -.6637283
      reg668 |          0  (omitted)
```

```
      reg669 |  -.3348251   .2315281    -1.45   0.148    -.7887951    .1191448
     smsa66r |   .2735886   .0880339     3.11   0.002     .1009756    .4462016
       _cons |   17.36837   .2406549    72.17   0.000      16.8965    17.84023
------------------------------------------------------------------------------

. predict res_ed76, residual

. reg lwage76  ed76 exper exper2 reg66* smsa66 res_ed76 , robust
note: reg668 omitted because of collinearity.

Linear regression                                      Number of obs     =      3,010
                                                       F(13, 2996)       =      82.16
                                                       Prob > F          =     0.0000
                                                       R-squared         =     0.2552
                                                       Root MSE          =     .38384

------------------------------------------------------------------------------
             |               Robust
     lwage76 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ed76 |   .1416412   .0556841     2.54   0.011     .0324583     .250824
       exper |   .1114847   .0246171     4.53   0.000     .0632165    .1597529
      exper2 |  -.2427267   .0330305    -7.35   0.000    -.3074915    -.177962
      reg661 |   .0850828   .0567551     1.50   0.134    -.0262001    .1963657
      reg662 |   .1884525   .0534306     3.53   0.000     .0836881     .293217
      reg663 |   .2248428   .0537804     4.18   0.000     .1193925    .3302931
      reg664 |   .1298129   .0534874     2.43   0.015     .0249371    .2346888
      reg665 |   .0621124   .0797695     0.78   0.436    -.0942962     .218521
      reg666 |   .0790861   .0840409     0.94   0.347    -.0856975    .2438698
      reg667 |   .0536368   .0762723     0.70   0.482    -.0959146    .2031883
      reg668 |          0  (omitted)
      reg669 |   .1946134   .0498494     3.90   0.000     .0968708    .2923559
     smsa66r |   .0803322   .0268919     2.99   0.003     .0276038    .1330606
    res_ed76 |  -.0589462   .0558638    -1.06   0.291    -.1684815    .0505892
       _cons |   3.445078   .9730124     3.54   0.000     1.537238    5.352918
------------------------------------------------------------------------------

. test res_ed76

 ( 1)  res_ed76 = 0
```

```
       F(  1,  2996) =    1.11
            Prob > F =    0.2914
```

7. Now estimate the model by GMM using `nearc4a` and `nearc4b` as instruments. What kind of test can you do here? Do this test. Are there meaningful differences here with 2SLS and GMM when using these two instruments?

Solution:

Here, we can do an overidentification test (i.e. a test based on the fact that we have more orthogonality conditions than parameters).

We cannot reject the fact that our orthogonality conditions are satisfied. If all our other assumptions are satisfied, this is an indication that our extra instrument(s) is (are) valid.

```
    . ivregress gmm lwage76  exper exper2 reg66* smsa66  (ed76  = nearc4a nearc4b )
note: reg669 omitted because of collinearity.

Instrumental variables GMM regression             Number of obs   =      3,010
                                                  Wald chi2(12)   =     417.93
                                                  Prob > chi2     =     0.0000
                                                  R-squared       =     0.0914
GMM weight matrix: Robust                         Root MSE        =     .42296

------------------------------------------------------------------------------
             |               Robust
     lwage76 | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ed76 |   .1737952   .0488537     3.56   0.000     .0780437    .2695466
       exper |   .1252862   .0221042     5.67   0.000     .0819629    .1686096
      exper2 |  -.2464224   .0384342    -6.41   0.000    -.3217521   -.1710927
      reg661 |  -.1046738   .0441579    -2.37   0.018    -.1912217   -.0181259
      reg662 |   .0026613   .0348601     0.08   0.939    -.0656633     .070986
      reg663 |   .0400523   .0349833     1.14   0.252    -.0285138    .1086183
      reg664 |  -.0609951   .0420444    -1.45   0.147    -.1434005    .0214104
```

```
      reg665 |  -.1029791   .0552501    -1.86   0.062    -.2112674    .0053091
      reg666 |  -.0841613   .0602088    -1.40   0.162    -.2021683    .0338457
      reg667 |   -.115256   .0527571    -2.18   0.029     -.218658   -.0118541
      reg668 |  -.2050749   .0541947    -3.78   0.000    -.3112945   -.0988553
      reg669 |          0  (omitted)
     smsa66r |   .0671115   .0254235     2.64   0.008     .0172824    .1169407
       _cons |    3.08733   .8397108     3.68   0.000     1.441527    4.733132
------------------------------------------------------------------------------
Instrumented: ed76
 Instruments: exper exper2 reg661 reg662 reg663 reg664 reg665 reg666 reg667
              reg668 smsa66r nearc4a nearc4b

. estat firststage

  First-stage regression summary statistics
  --------------------------------------------------------------------------
               |            Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.     F(2,2996)   Prob > F
  -------------+------------------------------------------------------------
          ed76 |  0.4592      0.4568       0.0063       10.0162    0.0000
  --------------------------------------------------------------------------


. estat overid

  Test of overidentifying restriction:

  Hansen's J chi2(1) =  .72488 (p = 0.3945)

.
end of do-file
```

Exercise 3

This exercise is based on the paper by Ginsburgh and van Ours (2003), Expert Opinion and Musical Competition, American Economic Review, 93(1), 289-296 that you can download here. After reading the introduction of the paper as well as section I and II you can replicate the paper and conduct a full blown IV analyzes (without reading the rest of the paper.) by answering the following questions. The dataset for this exercise is on Brightspace.

1. What is the main research question of the paper?

Solution: See the other pdf for the solution of this exercise

2. What are the main methodological challenges in order to answer this research questions? What is the main idea of the authors to solve it. Answer verbally and formally.
3. Please produce descriptive statistics of the data.
4. Run OLS regressions of number of critics on the variable negranking. How would you interpret the result?
5. Run first stage regressions for two different sets of instruments and test for relevance of the instruments.
6. Conduct a 2SLS estimation for the two sets of instruments using number of critics as outcome variable. What is your conclusion?
7. Conduct postestimation test after 2SLS concerning the relevance of the instrument, overidentifying restrictions and the endogeneity of regressor. What is your conclusion?
8. Now conduct a 2SLS in 2 separate steps, together with the Hausman test.

Exercise 4

We want to estimate the following equation on a cross-sectional sample of individuals:

$$\log oop_i = \beta_0 + \beta_1 \log income_i + \beta_2 1\{health_i = bad\} + \varepsilon_i$$

where oop_i measures out-of-pocket medical spending, $income_i$ is income and $1\{health_i = bad\}$ is an indicator variable equal to 1 if the health of the individual is bad and to 0 otherwise.

1. Given an interpretation for the OLS estimates of the coefficients β_1 and β_2 below.

```
. reg l_oop l_income bad_health

Source       |       SS           df       MS      Number of obs   =     2,408
-------------+----------------------------------   F(2, 2405)      =     71.17
Model        |   199.42561         2  99.7128051   Prob > F        =    0.0000
Residual     |   3369.5599     2,405  1.40106441   R-squared       =    0.0559
-------------+----------------------------------   Adj R-squared   =    0.0551
Total        |  3568.98551     2,407   1.4827526   Root MSE        =    1.1837

------------------------------------------------------------------------------
l_oop        | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
l_income     |   .4638982   .0410567    11.30   0.000     .3833881    .5444084
bad_health   |   .2763573     .04988     5.54   0.000      .178545    .3741696
```

```
_cons        |   3.668296   .3874609     9.47   0.000     2.908505    4.428088
------------------------------------------------------------------------------
```

Solution: An income higher by 1% is associated with out-of-pocket medical spending roughly .46% higher. Being in bad health compared to being in good health is associated with out-of-pocket medical spending about 27% higher. (IMPORTANT: here we used *associated with* and not terms like *generates* or *causes*. This is because the latter would imply that we claim that the regression measures causal relations. To do this, we would have to argue why there would not be any endogeneity in the above regression if the parameter we are interested in is a causal one.)

2. After running the OLS regression below, we run the following code. Why would we do that? What can you test based on these results? Give the result of this test.

```
predict resid_oop, residuals
gen resid_oop2 = resid_oop^2
gen l_income2 = l_income^2
gen l_income_bad = l_income*bad_health
reg resid_oop2 l_income bad_health l_income2 l_income_bad

Source |       SS           df       MS      Number of obs   =     2,408
-------------+----------------------------------   F(4, 2403)      =     28.23
Model        |  799.728396         4  199.932099   Prob > F        =    0.0000
Residual     |  17017.1947     2,403  7.08164574   R-squared       =    0.0449
-------------+----------------------------------   Adj R-squared   =    0.0433
Total        |  17816.9231     2,407  7.40212842   Root MSE        =    2.6611

------------------------------------------------------------------------------
resid_oop2   | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
l_income     |  -9.560433    1.44409    -6.62   0.000    -12.39222   -6.728642
bad_health   |  -2.169031   1.789421    -1.21   0.226    -5.677999    1.339938
l_income2    |   .4748066   .0763238     6.22   0.000     .3251394    .6244739
l_income_bad |   .2892066   .1921166     1.51   0.132    -.0875247     .665938
_cons        |   48.90203   6.851412     7.14   0.000     35.46674    62.33732
------------------------------------------------------------------------------
```

Solution: We want to test for (conditional) heteroskedasticity. If there is, we would need to adjust our standard errors and we could also trying using possibly more efficient estimators (feasible generalized least squares). The test we can do is a White test. We compute

$$nR^2 = 2408 \times 0.0559 = 134.61$$

We know that

$$nR^2 \sim \chi^2(4)$$

under the null of homoskedasticity. There is only 0.5% chances that a variable distributed as a $\chi^2(4)$ takes values larger than 14.86. Given the value of nR^2, we can therefore safely reject this null hypothesis.

3. After that we run the following code. `resid_oop2_pred` is the predicted value of the OLS residual u_i^2 as a function of the regressors. What type of estimation are we performing here and why? [Note: `reg y x [aweight=z]` is equivalent to doing an OLS regression of $\sqrt{z}y$ on $\sqrt{z}x$.]

```
. reg l_oop l_income bad_health [aweight=1/resid_oop2_pred]
(sum of wgt is 1,955.81755912769)

Source       |       SS           df       MS      Number of obs   =     2,408
-------------+----------------------------------   F(2, 2405)      =     60.86
Model        |  153.945099         2  76.9725494   Prob > F        =    0.0000
Residual     |   3041.8527     2,405  1.26480362   R-squared       =    0.0482
-------------+----------------------------------   Adj R-squared   =    0.0474
Total        |   3195.7978     2,407  1.32770993   Root MSE        =    1.1246

------------------------------------------------------------------------------
l_oop        | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
l_income     |   .3678196   .0374578     9.82   0.000     .2943666    .4412726
bad_health   |   .3171704   .0499936     6.34   0.000     .2191354    .4152054
_cons        |   4.559548   .3600055    12.67   0.000     3.853595    5.265501
------------------------------------------------------------------------------
```

Solution: This is a version of feasible GLS (more specifically weighted least squares). The reason we may want to do this is that, because there is heteroskedasticity, FGLS may possibly (although not with certainty) be more efficient than OLS.

4. Finally, we do the same regression but this time asking Stata to return heteroskedasticity-robust standard errors. Why would we want to do that?

```
. reg l_oop l_income bad_health [aweight=1/resid_oop2_pred], robust
(sum of wgt is 1,955.81755912769)

Linear regression                               Number of obs     =      2,408
F(2, 2405)        =      35.22
Prob > F          =     0.0000
R-squared         =     0.0482
Root MSE          =     1.1246

------------------------------------------------------------------------------
             |               Robust
l_oop        | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
l_income     |   .3678196   .0510678     7.20   0.000     .2676782     .467961
bad_health   |   .3171704   .0536404     5.91   0.000     .2119842    .4223566
_cons        |   4.559548   .4860953     9.38   0.000     3.606339    5.512757
------------------------------------------------------------------------------
```

Solution: FGLS will produce correct standard errors only if we have correctly modeled the form of the heteroskedasticity. As this is uncertain, it is good practice to use robust standard errors with FGLS in case some of the heteroskedasticity remains.

Exercise 4

Consider the following model:

$$y_i = \mathbf{z}_i'\boldsymbol{\beta} + \varepsilon_i$$

where:

- $\mathbf{z}_i' = (z_{1i}, ..., z_{Ki})$ with $\mathbb{E}\left((z_{1i}, ..., z_{K-1i})'\,\varepsilon_i\right) = \mathbf{0}$ and $\mathbb{E}\left(z_{Ki}\varepsilon_i\right) \neq 0$
- $\mathbf{x}_i' = (z_{1i}, ..., z_{K-1i}, x_{Ki})$ and $\mathbb{E}\left(\mathbf{x}_i\varepsilon_i\right) = \mathbf{0}$
- observations are i.i.d.
- $\mathbb{E}\left(\mathbf{x}_i\mathbf{x}_i'\right)$ exists and is invertible
- $\mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$ exists and is invertible

1. Why estimating the relation between y_i and $\mathbf{z}_i$ by OLS would produce an inconsistent estimate for $\boldsymbol{\beta}$?

   Because

$$\mathbb{E}\left(z_{Ki}\varepsilon_i\right) \neq 0 \Rightarrow \mathbb{E}\left(\mathbf{z}_i\varepsilon_i\right) \neq \mathbf{0}$$

2. Show that $\mathbb{E}\left(\mathbf{x}_i\varepsilon_i\right) = \mathbf{0} \Leftrightarrow \mathbb{E}\left(\mathbf{x}_i y_i\right) = \mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)\boldsymbol{\beta}$.

$$\mathbb{E}\left(\mathbf{x}_i\varepsilon_i\right) = \mathbf{0} \Leftrightarrow \mathbb{E}\left(\mathbf{x}_i\left(y_i - \mathbf{z}_i'\boldsymbol{\beta}\right)\right) = \mathbf{0} \Leftrightarrow \mathbb{E}\left(\mathbf{x}_i y_i\right) - \mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)\boldsymbol{\beta} = \mathbf{0} \Leftrightarrow \mathbb{E}\left(\mathbf{x}_i y_i\right) = \mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)\boldsymbol{\beta}$$

3 . What does this later condition suggests as an estimator for $\boldsymbol{\beta}$?

As $\mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$ is invertible, we have:

$$\boldsymbol{\beta} = \mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\mathbb{E}\left(\mathbf{x}_i y_i\right)$$

The estimator that this suggests is:

$$\hat{\boldsymbol{\beta}} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i y_i\right)$$

in which we just have replaced the expectations by their sample analogs.
4. Show that this estimator is consistent for $\boldsymbol{\beta}$.

$$\hat{\boldsymbol{\beta}} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i y_i\right)$$

$$\Rightarrow \hat{\boldsymbol{\beta}} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\left(\mathbf{z}_i'\boldsymbol{\beta} + \varepsilon_i\right)\right)$$

$$\Rightarrow \hat{\boldsymbol{\beta}} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\boldsymbol{\beta} + \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)$$

$$\Rightarrow \hat{\boldsymbol{\beta}} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)\boldsymbol{\beta} + \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)$$

$$\Rightarrow \hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)$$

$\{\mathbf{x}_i\mathbf{z}_i'\}$ is i.i.d. and $\mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$ so, by Kolmogorov's LLN:

$$\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right) \underset{p}{\rightarrow} \mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$$

and so as $\mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$ is invertible,

$$\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1} \underset{p}{\rightarrow} \mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)^{-1}$$

Similarly as $\mathbb{E}\left(\mathbf{x}_i\varepsilon_i\right) = \mathbf{0}$:

$$\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right) \underset{p}{\rightarrow} \mathbb{E}\left(\mathbf{x}_i\varepsilon_i\right) = \mathbf{0}$$

Hence

$$\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right) \underset{p}{\rightarrow} \mathbf{0}$$

And so

$$\hat{\boldsymbol{\beta}} \underset{p}{\rightarrow} \boldsymbol{\beta}$$

Here we have derived the IV estimator for $\boldsymbol{\beta}$ and proved its consitency. Importantly, this requires $\mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$ to be invertible. This condition requires that, when regressing the endogeneous variable z_{Ki} on BOTH the exogenous regressors $(z_{1i}, ..., z_{K-1i})$ and on the instrumental variable x_{Ki}, the regression coefficient on the instrumental variable x_{Ki} is different from 0. It also requires that the instrumental variable is orthogonal to the error term. In other words, we need to have $\mathbb{E}\left(x_{Ki}\varepsilon_i\right) = 0$.